

Grimdark Forge — Assessment Task 2 Summary Report

Course	COSC3046 Web Programming Studio
Assessment	Task 2 — Dynamic Web Application (Prototype)
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Repository	Same GitHub repository as Assessment Task 1, branch <code>main</code>
Hosted at	https://titan.csit.rmit.edu.au/~s4246254/welcome.php
Modules	Discussion Forum · Shopping Cart · Blog · Product Review & Rating

1. Updated group charter

The full charter is `CHARTER.md` in the repository. The changes since Assessment Task 1:

- **Shared site shell is now built server-side.** The header, navigation and footer are `shared/header.php`, `shared/nav.php` and `shared/footer.php`, pulled into every page with `require`. The path prefix for module folders is computed, not typed, so no page carries a copy of the shell.
- **Login/logout lives in the shell.** A username-only login (no password) on `account.php`; the header shows *Log in* or the account name plus *Log out* on every page. The username is kept in the PHP session and every module uses the same `current_user()` / `owns()` helpers for ownership.
- **Locale-switching mechanism confirmed as implemented.** The mechanism agreed in Assessment Task 1 — a header `<button>` that expands a list of locales, whole-site scope, stored against the session, reflected in `<html lang>` — was built exactly as described. Two locales are supported, `en-AU` and `de-DE`.
- **Working agreements updated** for the new constraints: raw PHP/JS only, no HTML validation attributes, no inline styles, rendered-page checks run in both locales before committing.

2. Test accounts

Log in from the **Account** link (or the *Log in* button in the header) by typing a username. No password is required.

Username	Display name	Sample content owned
kaya	Kaya Ellery	2 forum threads and replies, 2 blog posts, 2 reviews, a saved 3-item cart

Username	Display name	Sample content owned
toma	Toma Brandt	2 forum threads and replies, 1 blog post, 2 reviews
renn	Renn Vasco	1 forum thread and replies, 2 blog posts, 1 review
oksana	Oksana Reyes	1 forum thread and replies, 1 blog post, 2 reviews

Any other username (3–20 letters, digits or underscores) creates a new empty account.

Testing ownership. Log in as `kaya`, open the forum thread *Is the Sentinel actually worth its points at 1500?* — Kaya's reply carries *Your post* with Edit and Delete; Toma's opening post does not. Then log out and log in as `toma`: the controls move to Toma's post. Typing `forum/edit-post.php?id=1` while logged in as `kaya` returns a 403 page ("You cannot edit this post"), as do the equivalent blog, review and comment requests, so ownership is enforced by the server and not just hidden in the UI. The same pattern applies to blog posts and comments, reviews, cart lines and order confirmations (an order can only be viewed by the account that placed it).

Shared data. The store is server-wide: a thread started in one browser appears immediately in another, so two markers can interact (one posts, the other replies) and test ownership against each other. Guest carts are the only per-browser data.

Reset. *Reset demo data* on the account page returns every module to the seed content for all visitors. A web-server restart does the same.

3. Locales and how to test the toggle

Supported locales: **English (Australia)** — `en-AU` (default) and **German** — `de-DE`.

1. On any page, activate the globe button in the header (it reads *English* or *Deutsch*). It expands a list of the two locales; the current one is marked. The list also works with JavaScript disabled.
2. Choose *Deutsch (Deutschland)*. The page reloads with `<html lang="de-DE">`; the navigation, buttons ("Anmelden"/"Abmelden"), footer, headings, form labels, hints, validation messages and status messages are in German.
3. Check the formatting: prices read `42,00 AU$` rather than `$42.00`, dates read `13. August 2026 um 21:40` rather than `13 August 2026 at 9:40 pm`, and counts use German separators. The checkout and order-confirmation pages format every figure this way.
4. Navigate to another module — the locale is kept for the whole session.
5. Submit a form with an error (for example an empty username): the JavaScript messages are German too, and so are the PHP messages if JavaScript is disabled.
6. Choose *English (Australia)* to switch back.

User-written content (post bodies, review text, product descriptions) is not translated, as the brief allows at this stage.

4. Known issues and limitations

- **How the in-memory store survives between requests.** PHP forgets everything at the end of a request, so `shared/store.php` parks the associative arrays in APCu shared memory when the server has it, and otherwise in one serialised scratch file in the system temp directory (outside the web root, under an exclusive lock). The file backend is technically a file, but it holds nothing but the in-memory structure, is never queried, and is discarded on reset or clean-up — there is no database until Assessment Task 3.
- **Uploaded images are held in the store as data URIs**, capped at 1 MB each. Many large uploads would make every request load a larger store; library images avoid this.
- **PHP `intl` extension.** With `intl` present the server formats dates and prices with ICU; without it a built-in table is used for the two locales. In either case the browser re-formats with `Intl`, so the visible result is the same; only the no-JS fallback differs slightly (e.g. `08:22` with a leading zero).
- **Reply threading after deletion.** A deleted forum post is hidden and its replies are re-attached to the nearest visible ancestor, so a conversation can read oddly. The delete page warns about this before confirming.
- **Deleted forum posts are retained in memory** (as the brief requires) but there is no moderator view of the audit record yet; the record is only inspectable in the store.
- **No pagination controls appear** while the seed data fits on one page. Pagination is implemented (6 threads / 8 products / 6 posts / 6 reviews per page) and the controls appear once enough content is created.
- **The W3C validator and Lighthouse** have not been run against the hosted copy at the time of writing; the scripted checks cover the structural half of what they report.

5. Verification performed

- Every page (23 distinct pages) rendered through PHP in both locales and checked for the correct `<html lang>`, one `h1`, no heading jumps, no HTML validation attributes, no inline styles, no duplicate ids, `alt` on every image, a label for every control, every `aria-describedby` target present, every internal link resolving and no PHP warnings — all pass.
- Each module exercised end to end over HTTP: create, retrieve, update and delete; validation failures re-rendering the form with inline messages and a summary; and edit/delete requests for another account's content refused with HTTP 403.
- In a real browser: JavaScript validation blocks submission and shows messages in the active language; the locale menu opens, closes and switches the site; cart line totals and the subtotal update live in the locale's number format.
- Checkout arithmetic checked by hand: with `FORGE10` applied, $208.50 - 20.85 \text{ discount} = 187.65$ goods, free standard delivery over 120.00, $\text{GST} = \text{total} \div 11$.
- `php -l` on every file; no PHP 8-only syntax, so the site runs on PHP 7.4 and later.